

SPEECH & ACCENT RECOGNITION

Accent-Based Regional Origin Identification

A 2D-CNN that classifies Kerala vs. Tamil Nadu English accents from a few seconds of raw speech, trained on seven fused acoustic feature sets.

97.4%

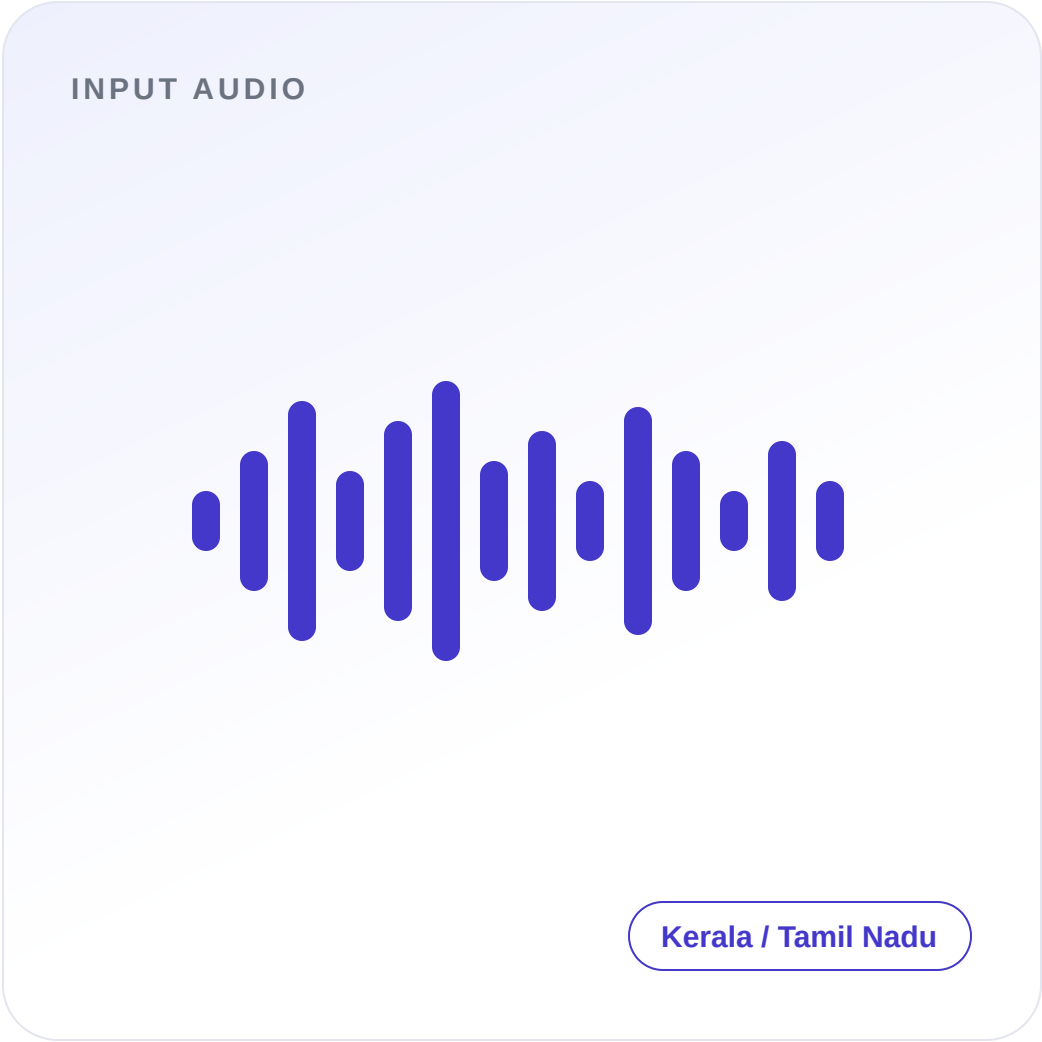
Test Accuracy

0.99

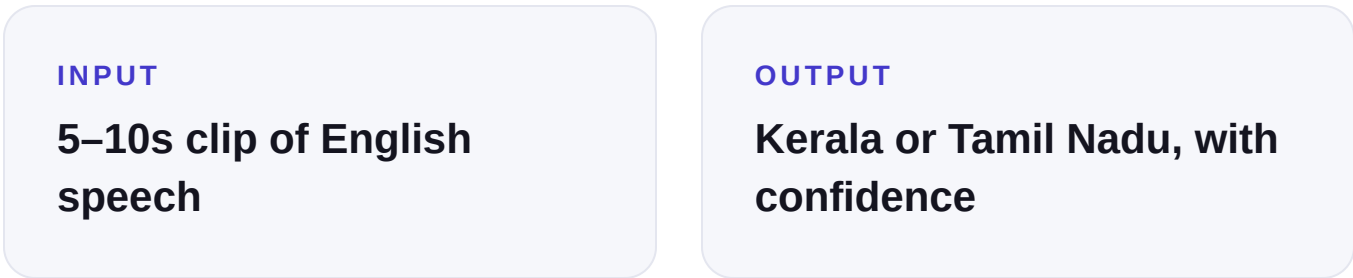
ROC – AUC

7

Fused Acoustic Features



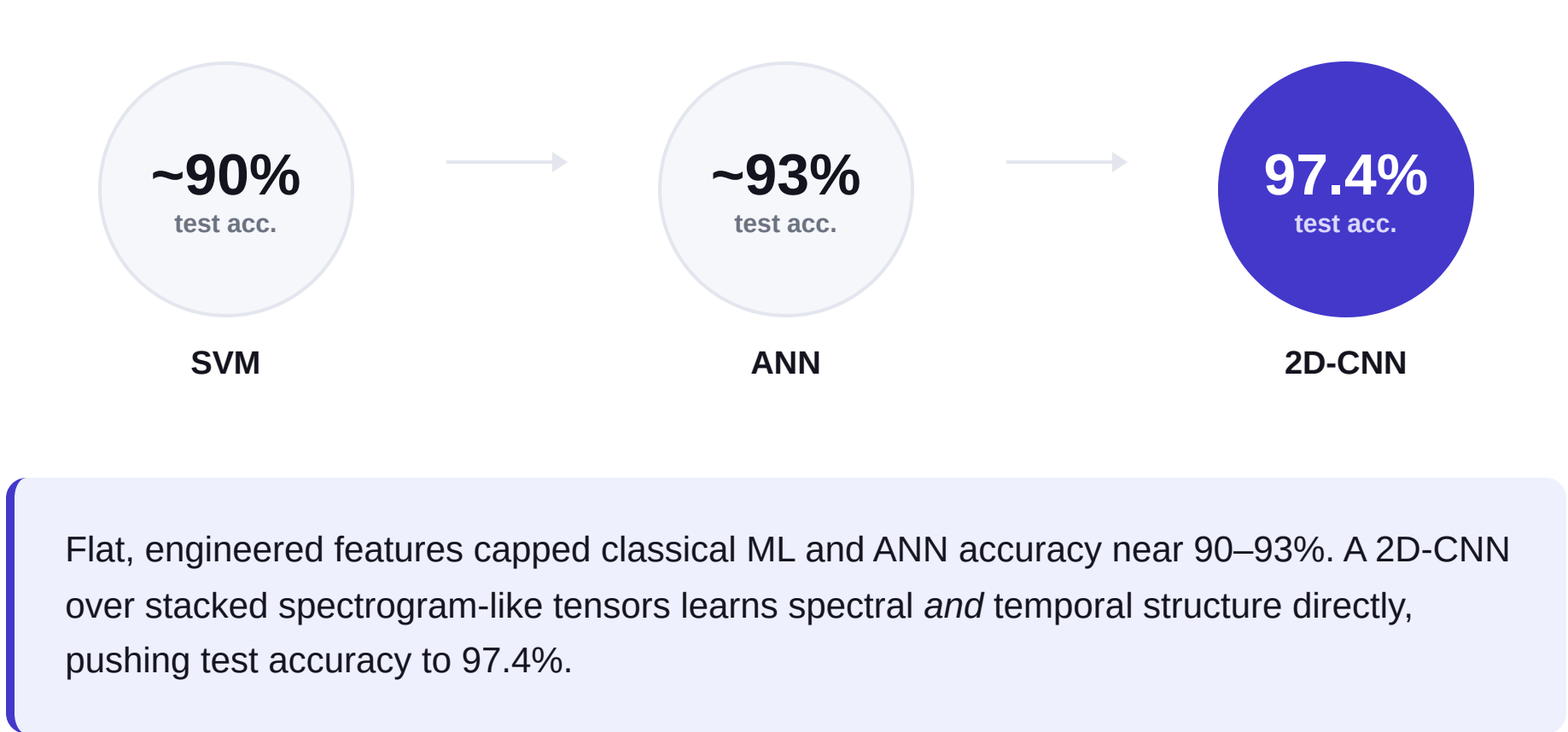
Problem & Approach

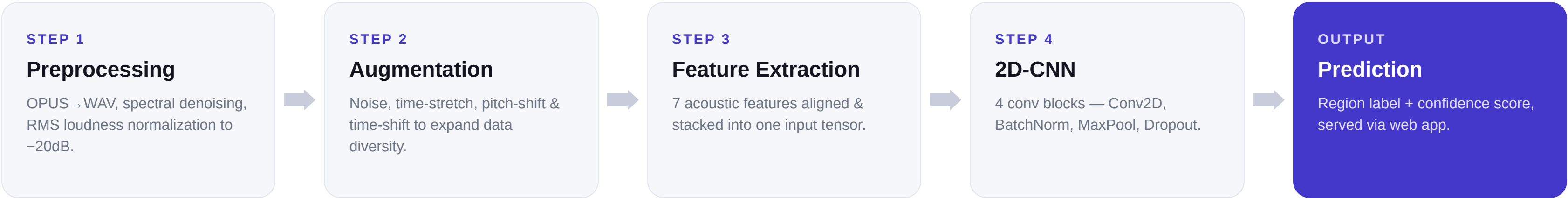


WHY IT'S HARD

- Kerala and Tamil Nadu English accents differ in subtle prosodic and phonetic cues, not obvious ones.
- Labeled regional speech data is scarce — the dataset was self-collected.
- Background noise and recording quality directly affect feature quality.

MODEL EVOLUTION





FUSED ACOUSTIC FEATURE SET



4

Convolutional blocks

50

Training epochs

80 / 20

Stratified train / test split

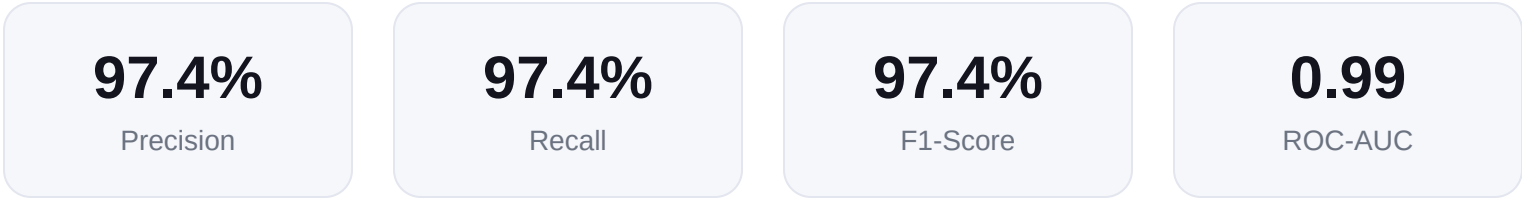
5-fold

Cross-validation

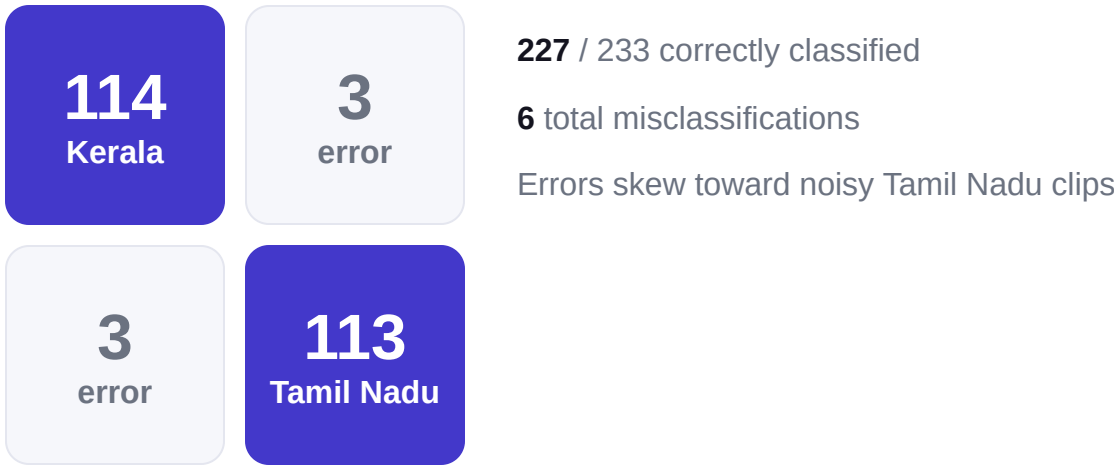
97.4%

Test-set accuracy, 2D-CNN model

Evaluated on a held-out, stratified 20% split — unseen speakers and recordings.



CONFUSION MATRIX — TEST SET (N = 233)



ACCURACY BY MODEL



MODELING

TensorFlow

Keras

scikit-learn

AUDIO PROCESSING

Librosa

noisereduce

ffmpeg

soundfile

DATA & UTILITIES

NumPy

pandas

Matplotlib

joblib

SERVING

Gradio

Hugging Face Spaces

HOW THE WEB APP WORKS

- 1

Upload or record
User provides a short WAV/MP3 clip of English speech in-browser.
- 2

Feature extraction + inference
Audio is processed and passed through the trained 2D-CNN.
- 3

Prediction returned
Predicted region, confidence %, and full probability breakdown.

Try the live classifier
Kerala vs. Tamil Nadu English Accent Classifier

GitHub Repo

Hugging Face Space